

NANYANG TECHNOLOGICAL UNIVERSITY**SEMESTER 1 EXAMINATION 2023-2024****SC3103/CE3103 – EMBEDDED PROGRAMMING**

Nov/Dec 2023

Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 4 pages.
2. Answer **ALL** questions.
3. This is a closed-book examination.
4. Questions **DO NOT** carry equal marks.
5. **Appendix A** at the end of the exam paper contains schedulability tests that you may find useful while answering some of the questions in this exam.

1. (a) Explain three key differences between the **Priority Ceiling Protocol (PCP)** and the **Stack Resource Policy (SRP)**.
(5 marks)
- (b) Consider a periodic task system defined as follows.
 - Tasks $\tau_1 = (12, 4, 11)$, $\tau_2 = (10, 3, 10)$, $\tau_3 = (9, 2, 7)$ and $\tau_4 = (6, 2, 6)$, where for a task $\tau_i = (T_i, C_i, D_i)$, T_i is the release period, C_i is the worst-case execution time and D_i is the relative deadline.
 - Suppose there are two shared resources R_1 and R_2 such that R_1 is used by τ_4 and τ_2 and R_2 is used by τ_3 and τ_1 .
 - Assume that each task will execute for 1 time unit and request for the shared resource at the end of that time unit. It will then hold the resource until it completes.
 - Suppose the first job of τ_1 is released at time 0, that of τ_2 is released at time 2, that of τ_3 is released at time 4 and that of τ_4 is released at time 5.

Note: Question 1 continues on page 2

Construct the schedule for the above system under **Rate Monotonic (RM) with PCP** for the time period $[0, 11]$. In your schedule, explain the decisions made by PCP.

(10 marks)

2. Consider a periodic real-time task system comprising 5 tasks scheduled on a **dual-core** processor. Suppose the tasks in the system are specified as $\tau_1 = (10, 3, 10)$, $\tau_2 = (6, 3, 6)$, $\tau_3 = (5, 3, 5)$, $\tau_4 = (5, 1, 5)$ and $\tau_5 = (3, 1, 3)$, where for a task $\tau_i = (T_i, C_i, D_i)$, T_i is the minimum release separation, C_i is the worst-case execution time and D_i is the relative deadline.

- (a) Is this task system feasible on a dual-core processor? Justify your answer.

(5 marks)

- (b) Is this task system schedulable under the **global RM** scheduling algorithm? Justify your answer.

(10 marks)

- (c) Suppose we wish to use partitioned scheduling strategies to schedule the above task system. While answering the below questions, show your workings in detail, including that of any schedulability tests employed.

- (i) Is there a partitioning strategy that would ensure schedulability under **EDF scheduling** on each core?

(8 marks)

- (ii) Is there a partitioning strategy that would ensure schedulability under **RM scheduling** on each core?

(12 marks)

3. (a) What is the Phony Target in GNU Make? Briefly describe its key characteristics and provide a common use case example in the Makefile.

(8 marks)

- (b) What is a zombie process? How does `fork()` relate to the creation of

Note: Question 3 continues on page 3

zombie processes? Explain how the parent process can handle zombie processes effectively.

(10 marks)

- (c) How many 'A', 'B' and 'C' are in the output of following code? Explain your answer.

```
int main()
{
    if (!fork()) {
        if (fork() || fork()) {
            printf("A\n");
        }
        else {
            printf("B\n");
        }
    }
    else {
        printf("A\n");
    }
    printf("C\n");
    return 0;
}
```

(7 marks)

4. (a) Briefly describe and compare anonymous pipe and named pipe.

(12 marks)

- (b) When we register an LKM with the kernel, what does device major number play its role? If the major number is zero, what does it mean?

(8 marks)

- (c) Briefly describe the primary bootloader in the boot process of embedded linux.

(5 marks)

Appendix A (Schedulability Tests)

In the following equations,

- n denotes total number of tasks in the system,
- m denotes number of cores,
- $\tau_i = (T_i, C_i, T_i)$ denotes a periodic/sporadic task with period T_i , execution time C_i , and relative deadline T_i ,
- B_i denotes maximum blocking time for task τ_i ,
- $HP(\tau_i)$ is the set of all tasks τ_k such that $T_k \leq T_i$, **excluding task τ_i** and
- n_i denotes total number of tasks in $HP(\tau_i)$.

1. **Feasibility test:** $\sum_{i=1}^n C_i/T_i \leq m$
2. **Sufficient test under RM:** $\sum_{i=1}^n C_i/T_i \leq n(2^{1/n} - 1)$
3. **Exact test under RM:** For each task τ_i , $R_i \leq T_i$, where R_i is the response time of task τ_i given by the solution of the following recursive equation
 - $R_i(0) = C_i$, and
 - $R_i(x) = C_i + \sum_{\tau_k: \tau_k \in HP(\tau_i)} \left\lceil \frac{R_i(x-1)}{T_k} \right\rceil C_k$
4. **Exact test with blocking under EDF:** For each task τ_i , $\sum_{\tau_k: \tau_k \in HP(\tau_i)} C_k/T_k + C_i/T_i + B_i/T_i \leq 1$
5. **Sufficient test with blocking under RM:** For each task τ_i , $\sum_{\tau_k: \tau_k \in HP(\tau_i)} C_k/T_k + C_i/T_i + B_i/T_i \leq (n_i + 1)(2^{1/(n_i+1)} - 1)$
6. **Exact test with blocking under RM:** For each task τ_i , $R_i \leq T_i$, where R_i is given by the solution of the following recursive equation
 - $R_i(0) = C_i + B_i$, and
 - $R_i(x) = C_i + B_i + \sum_{\tau_k: \tau_k \in HP(\tau_i)} \left\lceil \frac{R_i(x-1)}{T_k} \right\rceil C_k$

END OF PAPER

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Please read the following instructions carefully:

- 1. Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.